AI Gateway Evaluation Metrics: Reference Guide

This reference document provides concise definitions and evaluation frameworks for the metrics utilized in the AI Gateway performance dashboard. It serves as a standardized guide for analyzing the trade-offs between response quality, inference cost, and system latency.

1. Quality Metrics

Quality metrics ensure that the routed model meets semantic and correctness standards without sacrificing accuracy for cost or speed.

Metric	Definition & Operational Function	Measurement Method
Win Rate	The percentage of instances where the routed model's output is evaluated as superior or equal to a baseline premium model.	Automated LLM Judges or human A/B testing matrix.
Pref Accuracy	Measures how accurately the routing system predicts and aligns with specific user-defined preferences.	Percentage match against historical user feedback logs.
Judge Score	An automated, scalable score assigned to the output for evaluating contextual correctness, tone, and formatting.	Standardized 1-5 scale evaluated by advanced models (GPT-4, Claude).
Task Metrics	Traditional mathematical and linguistic evaluation scores applied to deterministic or specific NLP tasks.	F1-Score for classification, ROUGE for summarization, Exact Match for code.

2. Cost Metrics

Cost metrics track financial expenditures and optimize the resource footprint of LLM queries across cloud infrastructure.

Metric	Definition & Operational Function	Measurement Method
Cost Savings	The percentage reduction in overall API expenses achieved by routing to smaller/specialized models compared to using the premium baseline.	Calculated via cloud provider billing and API consumption logs.
Norm. Cost	The standard financial unit used to compare the cost-efficiency of routed traffic combinations.	Total monetary expenditure per 1,000 output tokens generated ($ / 1K tokens).
Router Overhead	The computational and operational cost incurred by running the routing model itself.	Monetary value of local infrastructure execution per request.

3. Latency Metrics

Latency metrics monitor the responsiveness and real-time user experience of the model endpoint interactions.

Metric	Definition & Operational Function	Measurement Method
TTFT	Time to First Token. The duration from prompt submission until the gateway receives the first character of the streaming response.	Recorded in milliseconds via Application Performance Monitoring (APM) tools.
TPOT	Time Per Output Token. Reflects the generation speed of subsequent tokens following the initial response burst.	Calculated as the total streaming duration divided by total output tokens (ms / token).
Router Latency	The inference time required by the gateway's proactive performance prediction module to make a routing decision.	Measured in milliseconds using server-side distributed tracing and telemetry.

4. Trade-Off & Optimization Metrics

These multi-dimensional frameworks assess how successfully the system balances competing objectives simultaneously.

Metric	Definition & Operational Function	Measurement Method
AUCQ	Area Under the Cost-Quality Curve. A comprehensive framework evaluating router efficiency across all budgetary levels.	Mathematical integration of quality scores plotted against varying budget constraints.
Cost @ X% Qual.	Determines the financial efficiency of the gateway when clamped to a specific quality constraint.	Calculating total expenditure threshold when quality satisfies X%.
Pareto Frontier	Identifies the set of routing configurations where no single metric can be improved without degrading another.	Multi-objective optimization algorithms plotted to map the optimal operational edge.